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Joint Optimization of Production Scheduling and Machine Switching Under Time-of-Use Electricity Tariffs

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Abstract

This paper investigates an energy-efficient single-machine scheduling problem under time-of-use (TOU) electricity tariffs with machine switching decisions. With the increasing importance of demand response programs in industrial systems, electricity cost can be reduced not only by shifting production to low-price periods but also by avoiding unnecessary energy consumption during idle times. To jointly exploit these two mechanisms, a mixed-integer linear programming (MILP) model is developed to integrate job scheduling and machine switching decisions within a unified framework. The model explicitly captures processing energy consumption, idle energy consumption, and switching-related costs under time-varying electricity prices. Computational experiments based on randomly generated instances demonstrate that the proposed model can effectively reduce total energy cost. Comparative results with a two-stage strategy show that the integrated optimization framework consistently achieves lower energy consumption. Sensitivity analysis under different TOU tariff settings further confirms that the performance advantage is influenced by electricity price variations but remains robust across different scenarios. Moreover, the benefit of machine shutdown becomes more pronounced as the scheduling horizon increases. These findings highlight the importance of jointly considering load shifting and machine switching in energy-aware production scheduling and provide practical insights for improving electricity utilization efficiency in industrial systems.

Keywords: time-of-use tariffs; production scheduling; machine switching; energy efficiency; mixed-integer linear programming; demand response

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1. Introduction

In modern society, energy plays a fundamental role in sustaining economic development and supporting human activities. Fossil fuels, as non-renewable energy resources, have long dominated the global energy supply and are expected to remain the primary energy sources in the foreseeable future. However, the continuous depletion of natural resource reserves and the relatively slow development of alternative energy technologies make it increasingly difficult to meet the rapidly growing energy demand. Meanwhile, energy production and consumption have been identified as the largest anthropogenic sources of greenhouse gas emissions and air pollutants, which significantly contribute to global climate change and environmental degradation. These challenges highlight the urgent need for improving energy efficiency and promoting more rational energy utilization.

Among different forms of energy, electricity has become the fastest-growing type of end-use energy consumption worldwide over the past decades. This trend reflects the increasing electrification of industrial processes and modern production systems. In particular, the industrial sector accounts for the largest share of delivered energy consumption and is also the dominant consumer of electricity globally [1]. Therefore, enhancing electricity utilization efficiency in industrial systems is of great significance for both economic performance and environmental sustainability. It also suggests that industrial systems should be regarded as a key domain for implementing advanced energy management strategies.

Demand response (DR) programs have been widely recognized as an effective approach to improving electricity utilization efficiency. DR refers to mechanisms that encourage end users to adjust their electricity consumption patterns in response to changes in electricity prices or incentive signals, thereby contributing to load balancing and system reliability. Unlike traditional demand-side management strategies that focus on reducing total energy consumption, DR mainly aims at reshaping the load curve through load shifting, transferring electricity usage from peak periods to off-peak periods [2]. Such mechanisms are particularly important because electricity generation during peak periods is typically more costly and less efficient due to the need for additional backup capacity.

DR programs can generally be categorized into price-based programs and incentive-based programs. Among them, time-of-use (TOU) tariffs, real-time pricing (RTP), and critical peak pricing (CPP) are the most representative price-based mechanisms [3–5]. In particular, TOU tariffs are widely adopted in industrial and commercial sectors due to their relatively simple implementation and stable pricing structure. Under TOU tariffs, electricity prices vary across predefined time intervals within a day, typically classified into peak, mid-peak, and off-peak periods. These pricing schemes reflect the varying costs of electricity generation and supply across different time periods and provide economic incentives for users to shift their electricity consumption.

However, it is important to note that DR programs primarily focus on load shifting rather than reducing total energy consumption. In practical production environments, machines may remain idle during certain periods due to scheduling decisions. If machines continue running during these idle periods, unnecessary energy consumption will occur. Therefore, relying solely on load shifting may not fully exploit the potential for improving energy efficiency.

To address this limitation, machine switching strategies, i.e., switching machines off during idle periods and restarting them when needed, have been proposed as an effective operational approach. When the idle period is sufficiently long, the energy saved by shutting down the machine can outweigh the additional energy and time costs associated with restarting it. Compared with other energy-saving measures, such as investing in energy-efficient equipment or adjusting processing speeds [6,7], machine switching strategies do not require additional capital investment, making them particularly attractive for practical applications.

Despite the increasing attention to both TOU-based scheduling and machine switching strategies, existing studies have largely treated these two aspects separately. Only limited research has attempted to integrate load shifting and machine switching within a unified optimization framework. As a result, there remains a gap in the literature regarding how to jointly optimize production scheduling and machine operational decisions under time-varying electricity prices.

Motivated by this observation, this paper investigates a single-machine scheduling problem under time-of-use electricity tariffs with machine switching decisions. The objective is to minimize the total energy cost by jointly optimizing job scheduling and machine

switching strategies. To address this problem, we develop a mixed-integer linear programming (MILP) model that captures both processing-related and switching-related energy consumption. Compared with existing studies, the proposed model provides a more flexible and realistic representation by allowing jobs to span multiple periods without restrictive assumptions on processing times, while explicitly capturing the trade-off between idle energy and switching cost. Computational experiments based on randomly generated instances demonstrate that the proposed approach can effectively reduce total energy costs and generate high-quality scheduling solutions under TOU tariffs.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the problem description and the proposed MILP model. Section 4 reports the results of numerical experiments, and Section 5 concludes the paper.

2. Literature Review

Varying electricity prices can be very effective in reshaping the load curve when end users are sensitive to price signals. This explains why such pricing schemes have been widely adopted by industrial customers, since manufacturers are always eager to seize every opportunity to reduce costs. Motivated by the desire to decrease electricity costs in production and to fulfill corporate social responsibility, demand response (DR) programs, especially time-of-use (TOU) tariffs, have attracted growing interest among researchers [8,9]. However, previous research on the application of DR programs has mainly focused on residential and commercial buildings, while manufacturing systems have been largely neglected [10]. In recent years, however, academic research, which increasingly combines theory with practical applications, has started to pay more attention to production scheduling problems under TOU tariffs.

Among studies on production scheduling problems that take TOU tariffs into account under different machine environments, single-machine scheduling problems have been preferred by many scholars. Fang et al. [11] considered a single-machine scheduling problem under TOU tariffs with the objective of minimizing total electricity cost. They separately studied two cases with different job processing speed settings, namely the uniform-speed case and the speed-scalable case. Approximation algorithms were proposed for the speed-scalable case. In their research, both preemptive and non-preemptive versions of the problem were considered, and the NP-hardness of the non-preemptive problem was proved. Inspired by their study of the preemptive version, Che et al. [12] developed a greedy insertion heuristic for the single-machine scheduling problem under TOU tariffs after proposing a continuous-time MILP model to minimize total electricity cost. In both of these studies, jobs are assigned in non-increasing order of their power consumption rates. Gong et al. [13] also formulated a state-based MILP model for an energy-cost-aware single-machine scheduling problem with a due-date constraint. They later extended their work and developed a generic algorithm to search for an energy-cost-effective schedule [14]. In the case studies of these two papers, both TOU tariffs and RTP were discussed for the single-machine scheduling problem. CPP was also considered in their earlier research. Shrouf et al. [15] considered a single-machine scheduling problem with varying electricity prices to minimize total energy consumption cost. They used a mathematical model to solve the problem and also proposed a genetic algorithm. The electricity pricing scheme in their research is closer to RTP, but the algorithm they proposed can be applied under both RTP and TOU tariffs. Zhang et al. [16] studied an energy-conscious single-machine scheduling problem under TOU tariffs to minimize total electricity cost. Wu et al. [17] investigated a single-machine scheduling problem with release dates under TOU tariffs. They proposed a period-based MILP model and developed a two-stage heuristic algorithm to solve the problem. Yuan et al. [18] studied single-machine scheduling with group

processing features, building a MILP model as well as a block-based dynamic programming algorithm. An improved iterative greedy algorithm is also introduced.

As for parallel-machine scheduling problems, Moon et al. [19] developed a new mathematical model and proposed a hybrid genetic algorithm for unrelated parallel-machines, with the aim of improving both production efficiency and energy efficiency. Ding et al. [7] also investigated an unrelated parallel-machine scheduling problem under TOU tariffs to minimize total electricity cost. In their study, the production deadline was predetermined and could not be exceeded. As for more complicated machine environments, Zhang et al. [20] attempted to minimize both electricity cost and carbon footprint for a flow shop under TOU tariffs. A bi-objective mixed-integer optimization model was established, and the trade-off between the two objectives was discussed. Luo et al. [21] studied a multi-objective hybrid flow shop scheduling problem in the presence of TOU tariffs, considering both production efficiency and energy efficiency. An ant colony optimization metaheuristic was proposed to investigate the problem. To minimize the total production cost in a flexible job shop, Moon and Park [22] built two models using constraint programming and mixed-integer programming, respectively, and compared the results obtained by the two models for the classic flexible job shop scheduling problem under TOU and CPP rates. Cheng et al. [23] investigated a uniform parallel batch machine scheduling problem, and solve it by formulating a MILP model. Feng et al. [24] studied identical parallel-machine scheduling with a two-stage time-of-use tariff and not-all-machine option. They formulated the problem into a min–max regret model and developed an iterative relaxation-based exact algorithm and a memetic differential evolution-based heuristic. There are also other studies on scheduling problems in manufacturing systems that take TOU tariffs into consideration [10,25–27], although the TOU tariff considered by Wang and Li [10] is based on summer and winter TOU rates.

Machine startup and shutdown have been regarded as a fast and easy way to achieve reductions in energy consumption. There are many studies that discuss this strategy in energy-conscious scheduling problems. Mouzon et al. [28] studied an energy consumption minimization problem for manufacturing equipment. They proposed several dispatching rules based on the idea that machines or equipment should be turned off when they are expected to remain idle for a period of time. A multi-objective mathematical model was also developed, with the objectives of minimizing energy consumption and total completion time. Mouzon and Yildirim [29] investigated a single-machine scheduling problem in which deciding when to turn the machine off and on, as well as how long it should remain powered off, is essential for minimizing energy consumption. Chen et al. [30] investigated how to reduce energy consumption through effective startup and shutdown of machines in production systems. Liu et al. [31] formulated a multi-objective optimization model with machine switching considerations, aiming to minimize both total carbon dioxide emissions and total completion time, and developed a non-dominated sorting genetic algorithm II after considering the properties of the model. Dai et al. [32] and Pach et al. [33] also incorporated machine switching into their research.

Although research on TOU tariffs and machine switch-off/on strategies has grown rapidly, only a few studies have combined TOU tariffs with machine switching. To the best of our knowledge, only Gong et al. [13,14] and Shrouf et al. [15] attempted to do so. However, Refs. [13,14] used a time-indexed formulation to build the model. In this way, binary decision variables are defined for each job and each time unit. As a result, the number of variables grows proportionally with the product of the number of jobs and the length of the planning horizon, which may lead to a substantial increase in model size. However, under TOU tariffs, electricity prices remain unchanged within each time period. This suggests that fine-grained time discretization may not always be necessary to capture

the structure of the problem. Decision variables are only needed at the job level and the tariff period level. If some jobs are processed entirely within one period, the electricity cost of these jobs is not affected by their sequence. In [15], the processing times of jobs and the machine switch-off/on times are assumed to be exact integer multiples of the periods, which does not conform well to reality. In addition, the setting in which all periods are of equal length is more similar to an RTP scheme than to a TOU tariff.

Motivated by the above observations, this paper studies a more general single-machine scheduling problem in which machine switching decisions are integrated with TOU-based electricity pricing. The problem considered in this work aims to jointly exploit two complementary mechanisms for cost reduction: load shifting across time periods with different electricity prices, and energy saving through shutting down the machine during sufficiently long idle intervals. In this sense, the proposed framework combines the advantages of demand response and equipment-level energy conservation. The comparison between our study and existing single-machine scheduling studies under TOU tariffs is presented in Table 1.

Table 1. Comparison of studies of single-machine scheduling under TOU tariffs.

Study	Machine Operation	Job Attributes	Solution Method
Fang et al. [11]	Speed scaling		MILP + exact algorithm
Che et al. [12]			MILP + greedy algorithm
Gong et al. [13]	Shutdown	Due date	MILP
Gong et al. [14]	Shutdown	Due date	MILP + genetic algorithm
Shrouf et al. [15]	Shutdown		MIP + genetic algorithm
Zhang et al. [16]			MILP
Wu et al. [17]		Release date	MILP + heuristic algorithm
Yuan et al. [18]		Group processing	MILP + greedy algorithm
This study	Shutdown		MILP

The main contribution of this study is therefore not merely the development of another scheduling model, but the establishment of a MILP formulation that explicitly links TOU tariffs, job allocation across periods, and machine startup/shutdown decisions in a unified optimization framework. Compared with [13–15], the proposed model better aligns with the piecewise-constant nature of TOU tariffs, and allows a more flexible shutdown and startup time. Moreover, the model is more compact in terms of modeling logic by avoiding explicit time discretization, which also helps reduce memory requirements in computation. Consequently, electricity cost can be reduced not only by using electricity during lower-price periods, but also by avoiding unnecessary idle energy consumption through appropriate machine switching decisions.

3. Problem Description and Mathematical Model

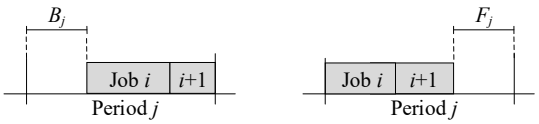
3.1. Problem Description

In the problem investigated in this study, a single machine is available to process a set of N jobs, all of which are ready for processing. The capacity of the machine is limited to processing at most one job at any given time. To ensure the quality of processing, preemption is not allowed throughout the entire production process. In addition, machine malfunction and maintenance are not considered in this study. Although the jobs to be processed are heterogeneous, they can be characterized by two key attributes, namely their processing time and their energy consumption rate per unit time. From a practical perspective, this single machine can represent a typical single-workstation operation in a manufacturing system, such as a grinding machine or CNC machine tools. The assumption that all jobs are available at time zero corresponds to short-term scheduling scenarios.

During the production process, the machine can be shut down when it is idle and restarted when needed. Each shutdown and reboot operation incurs a fixed amount of time and energy consumption. Therefore, energy cost arises from two main sources: the energy consumed during job processing and the energy consumed during machine switching operations. In the problem considered, the machine is powered by electricity, and the electricity price varies over time according to a TOU tariff. All jobs are required to be completed within a given time horizon. To guarantee the feasibility of the schedule, the length of the time horizon is predefined and must be greater than the total processing time of all jobs.

Based on the two sources of energy consumption described above, two key decision problems need to be addressed in order to minimize the total energy cost of the production process. The first problem is to assign jobs to periods with different electricity prices. This allocation problem involves determining in which period(s) each job is processed and how much processing time is assigned to each selected period. As a result, two types of decision variables are required to model this aspect, which is consistent with the modeling approach in [12]. The second key problem is to determine whether and when the machine should be shut down and restarted, since idle periods may arise due to the scheduling of jobs. In this study, each period is treated as a decision unit to determine whether the machine should be shut down at the end of that period. A binary decision variable is introduced for this purpose. It is worth noting that both time and energy considerations are essential when making shutdown decisions. Specifically, the length of idle periods, the time required for shutdown and reboot, as well as the energy consumption associated with idle running and switching operations must all be taken into account and compared.

To describe idle periods more precisely, two types of idle time are defined for each period, namely forward idle time and backward idle time. As illustrated in Figure 1, if there exists a period of idle time at the beginning of period j, before any assigned jobs start processing, this period is referred to as the backward idle time of period j. Similarly, if there exists a period of idle time at the end of period j, after all assigned jobs have been processed, this period is referred to as the forward idle time of period j. The lengths of both forward and backward idle times need to be determined within the model.



(**a**) (**b**)

Figure 1. (**a**) Illustration of backward idle time; (**b**) illustration of forward idle time.

Furthermore, if a job i is processed across two consecutive periods j and $j + 1$, this implies that there is no forward idle time in period j and no backward idle time in period $j + 1$. On the other hand, if a period is completely idle, repeated shutdown operations can be avoided. To capture these relationships, additional decision variables are introduced. Finally, a decision variable is defined to represent the energy cost associated with either idle running or machine switching at the end of each period.

3.2. Notations

The following notations summarize the indices and sets, parameters, and the decision variables for developing the MILP model:

Indices and sets

N: Number of jobs.

K: Number of periods.

Parameters

T: The minimum time needed for shutdown and reboot of the machine.

E: The energy consumption to start/reboot and shut down the machine.

p_i: Energy consumption required to process job i per unit time; $1 \leq i \leq N$.

p_{idle}: Energy consumption for running the machine idly per unit time.

t_i: Processing time of job i; $1 \leq i \leq N$.

d_j: The length of period j; $1 \leq j \leq K$.

c_j: The unit electricity price of period j; $1 \leq j \leq K$.

Decision variables:

x_{ij}: Assigned processing time of job i in period j; $1 \leq i \leq N, 1 \leq j \leq K$.

$$y_{ij} = \begin{cases} 1, \text{ if job } i \text{ or part of job } i \text{ is processed in period } j, \\ 0, \text{ otherwise.} \end{cases} ; 1 \leq i \leq N, 1 \leq j \leq K.$$

$$z_{ij} = \begin{cases} 1, \text{ if job } i \text{ is assigned both to period } j \text{ and } j+1, \\ 0, \text{ otherwise.} \end{cases} ; 1 \leq i \leq N, 1 \leq j \leq K-1.$$

$$U_j = \begin{cases} 1, \text{ if as least one job is processed in period } j, \\ 0, \text{ otherwise.} \end{cases} ; 1 \leq j \leq K.$$

$$W_j = \begin{cases} 1, \text{ if the machine is shut down in the back end of period } j, \\ 0, \text{ otherwise.} \end{cases} ; 1 \leq j \leq K-1.$$

B_j: The length of backward idle time of period j; $1 \leq j \leq K$.

F_j: The length of forward idle time of period j; $1 \leq j \leq K$.

T_j: The energy cost for running idly or switching the machine in the back end of period j; $1 \leq j \leq K-1$.

3.3. Mathematical Formulation

3.3.1. Objective Function

The objective of this study is to minimize the total energy cost over the entire planning horizon. This objective reflects both the cost of processing jobs under time-varying electricity prices and the cost associated with machine operation during idle periods.

Based on this objective, the mathematical formulation of the objective function is given in (1).

$$Obj = \text{Min} \sum_{i=1}^{N} \sum_{j=1}^{K} p_i c_j x_{ij} + \sum_{j=1}^{K-1} T_j - \sum_{j=1}^{K-1} (1 - U_j) E c_j \quad (1)$$

The objective function consists of three main components. First, the processing cost accounts for the electricity consumption of all jobs, weighted by the corresponding electricity prices in different periods. Second, the idle or switching cost captures the energy consumption incurred either when the machine runs idly or when it is shut down and restarted. Third, additional terms are included to avoid double counting in the case of consecutive idle periods.

To ensure the feasibility and logical consistency of the scheduling decisions, several groups of constraints are introduced in the model. These constraints are designed to capture five key aspects of the problem, including job assignment, inter-period relationships, time allocation and idle structure, idle period identification, and machine shutdown decisions.

3.3.2. Constraints

(a) Job assignment and consistency constraints

To ensure that each job is fully processed and properly assigned across different periods, a set of constraints is introduced to define the allocation of processing time and its consistency with assignment decisions. These constraints guarantee that the total assigned processing time of each job matches its required processing time and that the relationship between processing variables and binary assignment variables is logically consistent.

$$\sum_{j=1}^{K} x_{ij} = t_i, 1 \le i \le N; \tag{2}$$

$$x_{ij} \le t_i y_{ij}, 1 \le i \le N, 1 \le j \le K; \tag{3}$$

$$y_{ij} \le x_{ij}, 1 \le i \le N, 1 \le j \le K; \tag{4}$$

Constraint (2) ensures that if a job i is processed in multi-periods, the sum of its assigned processing times on each period is equal to its own processing time. Constraint (3) requires that when assigning job i to period j, the assigned processing time cannot exceed its processing time. Together with constraint (4), constraint (3) also ensures that y_{ij} gets the correct value corresponding to its definition. That is, if job i is assigned to period j, x_{ij} must be greater than 0, and then constraint (3) guarantees that y_{ij} gets the value of 1 since it is a binary, and if job i is not assigned to period j, then x_{ij} must be 0, which means the value of y_{ij} is 0 according to constraint (4).

(b) Inter-period relationship constraints

To capture the relationships of job assignments across adjacent periods and to ensure the continuity of job processing, a set of constraints is adopted to model whether a job spans multiple consecutive periods and to prevent infeasible or interrupted processing patterns as presented in [16]. The capacity limitation of the machine also needs to be reflected.

$$z_{ij} \ge y_{ij} + y_{i,j+1} - 1, 1 \le i \le N, 1 \le j \le K-1; \tag{5}$$

$$z_{ij} \le \left(y_{ij} + y_{i,j+1}\right)/2, 1 \le i \le N, 1 \le j \le K-1; \tag{6}$$

$$\sum_{i=1}^{N} z_{ij} \le 1, 1 \le j \le K-1; \tag{7}$$

$$\sum_{l=j+2}^{K} y_{il} \le K\left(1 - y_{ij} + y_{i,j+1}\right), 1 \le i \le N, 1 \le j \le K-2; \tag{8}$$

Constraints (5) and (6) assure the correctness when obtaining the value of z_{ij}. Constraint (5) presents that only when job i is assigned to both period j and $j + 1$ (i.e., $y_{ij} = 1$ and $y_{i,j+1} = 1$) can z_{ij} get the value of 1. And constraint (6) implies that as long as job i is absent from either period j or $j + 1$, maybe both, this job is not processed in these two adjacent periods, then the value of z_{ij} is 0. It should be noted that when considering jobs processed in adjacent periods, there is no need to take the last period into account because we assume that all the jobs are completed before the last period ends. Limited by the capacity of the machine, only one job can be processed at a time. This implies that for each pair of adjacent periods, only one job can be processed in them both, which is formulated in constraint (7). Constraint (8) ensures that jobs are processed uninterrupted. This constraint presents that when job i is assigned to period j but not to period $j + 1$ (i.e., $y_{ij} = 1$ and $y_{i,j+1} = 0$), then it must not be assigned to the periods after $j + 1$. In other occasions, this constraint is redundant.

(c) Time allocation and idle time constraints

To characterize the structure of each period, including both processing time and idle time, a set of constraints is introduced to ensure the proper allocation of time within each period. Some constraints are needed to define the relationship between processing time, forward idle time, and backward idle time, and ensure that the total time within each period is properly balanced.

$$x_{ij} \geq (y_{i,j-1} + y_{i,j+1} - 1)d_j, 1 \leq i \leq N, 2 \leq j \leq K-1; \tag{9}$$

$$d_j - \sum_{i=1}^{N} x_{ij} = B_j + F_j, 1 \leq j \leq K; \tag{10}$$

$$F_j \leq d_j(1 - \sum_{i=1}^{N} z_{ij}), 1 \leq j \leq K-1; \tag{11}$$

$$B_{j+1} \leq d_{j+1}(1 - \sum_{i=1}^{N} z_{ij}), 1 \leq j \leq K-1; \tag{12}$$

Constraint (9) guarantees that if job i is processed continuously from period $j-1$ to $j+1$, its assigned processing time in period j is equal to the length of this period. Constraint (10) requires that the sum of forward idle time and backward idle time in period j is equal to its length subtracting the total assigned processing time of jobs assigned to this period. Since the right side of the equation must not be smaller than 0, here exists a hint that all the assigned processing time of job i on period j, as well as their sum, will not exceed the length of this period. If there is a job i that is processed in both period j and $j+1$ (i.e., $z_{ij} = 1$), constraints (11) and (12) ensure that no forward idle time exists in period j and no backward idle time exists in period $j+1$. Conversely, if no job crosses the boundary of period j and $j+1$ when processing (i.e., $z_{ij} = 0$), the forward idle time of period j and backward idle time in period $j+1$ may exist but cannot exceed the length of their period respectively.

(d) Idle period identification and shutdown logic

To determine whether a period is idle and whether the machine should be shut down, a set of constraints is introduced to identify idle periods and link them with machine operation decisions. Here are some constraints to ensure that the machine is shut down when a period is completely idle and remains active when processing continuity must be maintained.

$$U_j \geq y_{ij}, 1 \leq i \leq N; 1 \leq j \leq K; \tag{13}$$

$$U_j \leq \sum_{i=1}^{N} y_{ij}, 1 \leq j \leq K; \tag{14}$$

$$W_j \geq 1 - U_j, 1 \leq j \leq K-1; \tag{15}$$

$$W_j \leq 1 - \sum_{i=1}^{N} z_{ij}, 1 \leq j \leq K-1; \tag{16}$$

According to the notations, the binary U_j is defined to judge if a period j is idle. Constraint (13) and (14) assure that U_j always gets the correct value. As long as there exists at least one job is assigned to period j, this period will not be idle, as is shown in constraint (13), but only if no job is assigned to period j, can it be told to be idle, as is shown in constraint (14). Constraint (15) requires that once period j is idle, the machine must be shut down. Constraint (16) states that for a period j, if a job is assigned both to it and its adjacent period $j+1$, the machine will not be shut down in period j due to the continuity of processing jobs.

(e) Shutdown feasibility and energy trade-off constraints

To ensure that machine shutdown decisions are both feasible and economically justified, a set of constraints is introduced to model the time and energy conditions associated with shutdown operations. These constraints compare idle running with shutdown-and-restart operations and ensure that shutdown is performed only when it leads to net energy savings.

$$F_j + B_{j+1} \geq W_j T, 1 \leq j \leq K-1; \tag{17}$$

$$(F_j c_j + B_{j+1} c_{j+1}) p_{idle} \geq W_j E c_j, 1 \leq j \leq K-1; \tag{18}$$

$$T_j \geq F_j c_j p_{idle} + B_{j+1} c_{j+1} p_{idle} - W_j (d_j c_j p_{idle} + d_{j+1} c_{j+1} p_{idle}), 1 \leq j \leq K-1; \tag{19}$$

$$T_j \geq W_j E c_j, 1 \leq j \leq K-1; \tag{20}$$

$$B_j \geq (1 - U_j) d_j / 2, 1 \leq j \leq K; \tag{21}$$

$$F_j \geq (1 - U_j) d_j / 2, 1 \leq j \leq K. \tag{22}$$

Constraint (17) and (18) present the two essential conditions for shutting down the machine, which are respectively time boundary and energy cost boundary. One essential condition presented in constraint (17) shows that for any pair of adjacent periods j and $j + 1$, if the forward idle time of period j, together with the backward idle time of period $j + 1$, compose an idle time interval that is longer than shutdown time boundary T, then the machine can be shut down. In other words, if the machine is to be shut down in period j, the above-mentioned time interval must exceed the time boundary T which can give enough time for a switch. Another essential condition presented in constraint (18) requires that to shut down the machine, the energy cost for running the machine idly in the idle time interval must exceed the energy cost Ec_j needed to shut down and reboot the machine, or else shutting down the machine will not be economical. Note that here we calculate the cost using the electricity price of the period when the shutdown happens. Constraints (19) and (20) ensure that either the machine is idle or shut down at the end of each period, and their energy cost can be calculated accurately. If the back end of period j is idle but not shut down (i.e., $W_j = 0$), constraint (19) will work and constraint (20) is redundant. On the contrary, if the machine is shut down in the back end of period j (i.e., $W_j = 1$), then constraint (20) comes into play leaving constraint (19) redundant. In order to guarantee the accuracy when judging whether to shut down the machine in the back end of the period, we enforce that the forward and backward idle time are equally split if the whole period is idle, which is formulated in constraints (21) and (22). It is worth noting that the equal-splitting rule implicitly assumes that the period is long enough that half of the length is no smaller than the minimum shutdown duration.

The proposed modeling framework focuses on a basic single-machine setting in order to highlight the interaction between scheduling decisions and machine switching decisions. More complex scheduling features, such as release dates and due dates, can be incorporated into the proposed model when defined at the level of TOU pricing periods. For more refined temporal representations, time-discretized formulations may be required. Such extensions would inevitably increase the complexity of the model.

4. Numerical Experiments

To evaluate the effectiveness of the proposed model, a series of computational experiments are conducted based on randomly generated instances. The experiments are designed to analyze how time-of-use (TOU) electricity pricing and machine switching decisions jointly influence the total energy cost under different scheduling conditions. The MILP model is solved by the commercial solver CPLEX (version 12.10.0.0 for 64-bit Windows).

4.1. Instance Generation and Parameter Settings

A set of computational experiments is conducted based on randomly generated instances. The purpose of these experiments is to examine the impact of time-of-use (TOU) electricity pricing and machine switching decisions on energy-aware production scheduling under different problem scales and levels of scheduling flexibility.

Three groups of instances are considered, with the number of jobs set to N = 20, 30, and 40, respectively. For each instance, the processing time of each job is independently generated from a uniform distribution over the interval [30, 120] min. This range is selected to represent a moderate level of heterogeneity in job durations and to ensure that the scheduling problem contains sufficient variation across jobs. In addition, the energy consumption rate of each job is randomly generated from [1, 5] unit(s), reflecting the fact that different jobs may impose different electricity demands during processing.

The electricity price follows the industrial and commercial TOU tariff structure adopted in Shaanxi Province, China. Specifically, one day is divided into off-peak, mid-peak, and peak periods. Since the objective of this study is to investigate the scheduling effect under realistic TOU pricing rather than to focus on a specific billing level, the tariff structure is introduced according to the corresponding time partitions. The detailed TOU periods are summarized in Table 2.

Table 2. Time-of-use electricity periods of Shaanxi Province, China.

Period Type	Time Interval	Price (Unit)
Off-peak	0:00–6:00	0.3
Mid-peak	6:00–11:00	1
Off-peak	11:00–14:00	0.3
Mid-peak	14:00–16:00	1
Peak	16:00–23:00	1.7
Mid-peak	23:00–24:00	1

This pricing structure provides differentiated electricity cost signals over the planning horizon and creates the possibility of shifting jobs from high-price periods to lower-price periods. As a result, it is suitable for testing whether the proposed model can effectively exploit TOU tariffs to reduce total energy cost.

Regarding machine switching, the minimum time required for shutting down and restarting the machine is set to 30 min, and the corresponding energy consumption for each shutdown-and-restart operation is set to 50 units. These two parameters jointly determine whether a shutdown decision is beneficial. More specifically, a shutdown is worthwhile only when the idle interval is long enough and the energy saved from avoiding idle running exceeds the energy consumed by the shutdown-and-restart operation.

To ensure feasibility, the total scheduling horizon must be greater than the sum of the processing times of all jobs. In order to further investigate the impact of schedule flexibility, a scaling coefficient is introduced to enlarge the total time horizon. Specifically, the length of the scheduling horizon is set to 1.2, 1.4, and 1.8 times the total processing time of all jobs, respectively. These three coefficients represent different levels of scheduling tightness. When the coefficient is relatively small, the schedule is tight and the room for load shifting or machine shutdown is limited. As the coefficient increases, the schedule becomes looser, allowing more opportunities for transferring jobs to low-price periods and for creating sufficiently long idle intervals that may justify machine shutdown.

The computational results are analyzed from the perspectives of solution quality and the effects of parameter settings. In particular, attention is given to how the job scale and the horizon scaling coefficient influence the total energy cost and the machine switching behavior.

4.2. Computational Results

In this section, the computational experiments are organized into three parts to systematically evaluate the performance of the proposed model. The first part focuses on the direct solution results under different problem scales and levels of scheduling flexibility. The second part is designed to evaluate the effectiveness of machine switching decisions by comparing the proposed model with cases in which the machine is not shut down. The third part presents a comparison with a two-step strategy, in which all jobs are first scheduled, and then shutdown decisions are made based on the length of idle periods and the associated energy consumption.

4.2.1. Experiments Under Different Problem Scales and Scheduling Flexibility

In the first set of experiments, we solve the proposed model directly under different combinations of job sizes and horizon scaling coefficients. For each combination of problem size and scaling coefficient, the model is solved using CPLEX. The resulting total energy cost and computational performance are recorded. This part of the experiment aims to observe how the model behaves under different levels of scheduling tightness and problem scales. The specific results are presented in Table 3, in which each instance is labeled according to its configuration. For example, instance 20-1 corresponds to the first instance with 20 jobs. *EC* represents the optimal electricity cost obtained by solving the model with a subscript to distinguish different scheduling horizons.

Table 3 lists the detailed results for all the tested instances mainly to illustrate the correlation between total power consumption and problem size, the correlation between total power consumption and scheduling cycle length, and the correlation between computation time and problem size. We can see from the results that both the problem scale and the scheduling horizon have impact on the solution performance.

First, the impact of problem scale on both solution quality and computational performance is significant. As the number of jobs increases from 20 to 40, the overall energy cost naturally increases due to the larger problem size. More importantly, the computational time grows dramatically. For 20-job instances, the average computational time is generally below 2 s, indicating that the model can be solved very efficiently. For 30-job instances, the computational time increases to a range of several seconds to around 50 s. When the problem size reaches 40 jobs, the computational time rises sharply, often reaching hundreds or even over one thousand seconds. This result clearly demonstrates that the problem becomes computationally challenging as the scale increases, which is consistent with the combinatorial nature of the model. For those 40-job instances that could not be solved to proven optimality within the 1800 s time limit, the relative optimality gaps at termination range from 0.01% to 0.2%. Although the obtained solutions are reasonably close to optimal, closing the remaining gaps is time-consuming, indicating slow convergence for some larger instances.

Second, the impact of scheduling horizon on *EC* is clearly observed. For all instance sizes, the energy cost consistently decreases as the scheduling horizon increases from 1.2 to 1.8 times of total processing time. This trend indicates that a larger scheduling horizon provides more flexibility for job allocation, allowing the model to better exploit low electricity price periods and thus reduce the total energy cost. This pattern is particularly evident in small- and medium-scale instances (20 and 30 jobs), where $EC_{1.8}$ is almost always significantly lower than $EC_{1.2}$. In most cases, the cost reduction is substantial, demonstrating the effectiveness of introducing temporal flexibility into the scheduling process. However, for some instances (e.g., instance 20-4, 30-4), the energy cost remains unchanged between $EC_{1.2}$ and $EC_{1.4}$. This suggests that, for these instances, the additional

flexibility provided by a slightly larger horizon is not sufficient to alter the optimal schedule, and the solution is already locally stable.

Table 3. Results of the instances with different numbers of jobs and scheduling horizons.

Instance	$EC_{1.2}$	$EC_{1.4}$	$EC_{1.8}$	Computational Time (s)
20-1	1978.3	1678.7	1356.9	0.60
20-2	2588.6	2193.4	1704.7	1.54
20-3	2070.5	1998.4	1698.0	1.65
20-4	1781.1	1781.1	1509.4	0.66
20-5	2241.2	2138.3	1916.3	0.81
20-6	2898.6	2500.6	1927.9	2.92
20-7	1643.8	1555.6	1379.8	1.09
20-8	1938.0	1701.0	1701.7	10.57
20-9	2528.3	2155.5	1712.3	1.93
20-10	2185.6	1909.4	1910.1	1.17
30-1	4323.1	3740.7	3224.0	38.03
30-2	4237.2	4153.9	3154.5	7.41
30-3	5054.9	3884.8	3501.8	15.38
30-4	3171.4	3171.4	2274.2	13.65
30-5	2989.9	2375.6	2240.4	6.26
30-6	3109.0	3026.4	2483.4	22.73
30-7	3252.1	2542.6	2285.6	11.53
30-8	3468.1	2683.7	2426.7	8.55
30-9	3818.7	2871.2	2614.2	49.63
30-10	4410.7	3377.8	3005.3	52.66
40-1	*	5711.8	*	1206.21
40-2	5079.1	*	3932.0	672.34
40-3	5578.6	5505.4	4280.3	75.99
40-4	3955.0	3593.1	3118.4	40.77
40-5	3838.3	3762.3	3021.6	252.10
40-6	3964.1	3350.5	*	612.51
40-7	4752.3	4676.3	3691.3	135.65
40-8	4480.9	3962.2	3181.2	167.84
40-9	4045.8	3961.8	3228.4	465.89
40-10	4745.3	4675.3	*	646.03

* Optimal solution is not obtained within 1800 s.

4.2.2. Comparison with the No-Shutdown Situation

In the second set of experiments, we evaluate the contribution of machine switching decisions by comparing the proposed model with a benchmark scenario in which machine shutdown is not allowed. That is, the machine is assumed to remain active throughout the scheduling horizon.

We use the same set of instances as in the first set of experiments. Under this setting, two models are solved: the proposed model, which allows machine shutdown decisions and a benchmark model, in which the machine is not allowed to shut down and must remain active during idle periods. The benchmark model is obtained by fixing $Wj = 0$ for all j in the proposed formulation. In addition, the objective function is revised by removing the second and third components and incorporating an additional term $\sum_{j=1}^{K} (B_j + F_j) p_{idle} c_j$ to explicitly capture the energy cost during idle periods.

For each instance, the total energy cost under the without machine shutdown condition is calculated. Based on these results, the relative energy-saving ratio (denoted as gap) achieved by allowing machine shutdown is calculated as:

$$Gap = \frac{EC' - EC}{EC'} \tag{23}$$

where EC' represents the electricity cost of the no-shutdown model. The results are presented in Table 4. To enhance readability and highlight the trends, Table 4 reports only the gap under different scheduling horizons, while the specific EC' values are omitted.

Table 4. Results of instances with and without shutdown.

Number of Jobs	***20***			**30**			**40**		
Instance	$Gap_{1.2}$	$Gap_{1.4}$	$Gap_{1.8}$	$Gap_{1.2}$	$Gap_{1.4}$	$Gap_{1.8}$	$Gap_{1.2}$	$Gap_{1.4}$	$Gap_{1.8}$
1	24.25%	36.88%	60.41%	18.81%	31.07%	47.44%	**	29.56%	**
2	22.34%	30.04%	55.96%	23.43%	28.79%	49.33%	16.80%	**	48.37%
3	21.70%	32.13%	53.20%	14.65%	27.63%	44.45%	15.07%	26.23%	46.51%
4	32.61%	32.61%	46.50%	29.69%	29.69%	49.68%	23.71%	39.58%	49.40%
5	18.21%	29.67%	49.60%	22.29%	38.27%	55.52%	20.69%	34.38%	55.24%
6	19.02%	28.59%	52.20%	29.44%	35.70%	55.28%	21.88%	41.27%	**
7	24.55%	37.24%	58.02%	22.82%	37.69%	55.56%	17.11%	29.44%	50.18%
8	30.21%	37.12%	51.92%	21.07%	36.11%	53.89%	16.85%	34.58%	54.78%
9	22.50%	30.23%	55.78%	19.19%	34.41%	51.94%	21.66%	33.96%	54.05%
10	28.26%	34.95%	49.33%	18.64%	31.82%	48.98%	19.59%	30.59%	**
Average	**24.37%**	**32.95%**	**53.29%**	**22.00%**	**33.12%**	**51.21%**	**19.26%**	**33.29%**	**51.22%**

** Cannot obtain a gap due to lack of optimal solution.

The results in Table 4 clearly demonstrate the effectiveness of incorporating machine shutdown decisions. Across all instances and parameter settings, the proposed model consistently achieves lower energy costs compared with the no-shutdown benchmark, leading to significant energy savings.

The results exhibit a highly consistent pattern across all instance sizes. For N = 20, 30, and 40, the average energy-saving ratio is around 20%, 30%, and 50% with a scheduling horizon equal to 1.2, 1.4, and 1.8 times of total processing time. This indicates that the proposed model achieves substantial and stable energy savings under all settings.

The underlying reason lies in the interaction between idle energy and scheduling decisions. In the no-shutdown scenario, machines must remain active during idle periods. To reduce idle energy consumption, the model may shift some processing operations into higher-price (peak) periods in order to maintain a more compact schedule. As the scheduling horizon expands, this trade-off becomes more pronounced, leading to increased energy costs in the no-shutdown model. In contrast, the proposed model can shut down machines during idle intervals, thereby avoiding unnecessary energy consumption. As a result, the advantage of allowing shutdown becomes more significant as the horizon increases.

4.2.3. Comparison with Shutdown After Scheduling

In this set of experiments, we compare the proposed joint optimization model with a more practical and intuitive two-step strategy, in which job scheduling is first determined, and the shutdown strategy is then applied afterward based on the resulting idle intervals.

The purpose of this comparison is to evaluate the benefit of integrating scheduling and machine switching decisions within a unified optimization framework. To ensure a fair comparison, the instances are under identical parameter settings with the former two sets of experiments. The calculation of the gap is the same as that of the previous set. The

detailed results are reported in Table 5. The specific electricity cost values, which can be deduced by using the gap, are also omitted in Table 5.

Table 5. Results of the proposed model and the two-step strategy.

Number of Jobs	***20***			**30**			**40**		
Instance	$Gap_{1.2}$	$Gap_{1.4}$	$Gap_{1.8}$	$Gap_{1.2}$	$Gap_{1.4}$	$Gap_{1.8}$	$Gap_{1.2}$	$Gap_{1.4}$	$Gap_{1.8}$
1	6.14%	11.53%	17.03%	2.83%	8.20%	12.31%	**	5.86%	**
2	4.04%	5.01%	15.03%	1.86%	4.63%	8.71%	6.78%	**	11.70%
3	3.08%	10.90%	7.95%	0.69%	5.67%	9.71%	2.54%	3.12%	11.50%
4	4.01%	4.01%	4.77%	5.51%	5.51%	11.32%	9.27%	14.14%	10.04%
5	3.21%	7.49%	6.20%	7.03%	19.70%	16.01%	7.64%	11.43%	14.20%
6	2.79%	2.35%	13.64%	11.88%	9.11%	23.18%	11.49%	17.17%	**
7	4.87%	11.24%	11.04%	11.16%	14.25%	15.75%	4.80%	7.58%	12.83%
8	9.48%	9.13%	5.88%	6.08%	6.84%	16.22%	3.11%	9.79%	14.64%
9	4.19%	4.62%	12.37%	4.84%	15.38%	16.34%	9.52%	12.11%	16.81%
10	4.78%	8.78%	9.71%	1.43%	4.80%	9.98%	7.93%	7.64%	**
Average	**4.66%**	**7.51%**	**10.36%**	**5.33%**	**9.41%**	**13.95%**	**7.01%**	**7.64%**	**13.10%**

** Cannot obtain a gap due to lack of optimal solution.

The results show that the proposed model outperforms the two-step strategy across all tested instances. Compared with the two-step strategy, the proposed model achieves additional cost savings ranging from approximately 4% to 14%, depending on the problem size and the time horizon. It highlights the importance of jointly optimizing scheduling and machine switching decisions. In the two-step strategy, job scheduling is determined without considering future shutdown opportunities, which leads to suboptimal idle patterns and limits the effectiveness of subsequent shutdown decisions. In contrast, the proposed model explicitly coordinates these decisions, thereby enabling more effective utilization of idle intervals and better alignment with electricity price variations.

Overall, the results confirm that machine shutdown decisions play a crucial role in improving energy efficiency under TOU tariffs. The proposed model not only reduces total energy cost but also effectively exploits scheduling flexibility to avoid excessive idle energy consumption. This indicates that machine switching becomes more valuable in flexible scheduling environments.

4.3. Sensitivity Analysis

As electricity price differentials are the primary drivers of shutdown decisions, it is important to evaluate how variations in tariff structures affect the performance of the proposed approach. To further examine the robustness of the proposed model with respect to electricity pricing, we conduct a sensitivity analysis on the TOU tariff levels.

In this experiment, we consider instances with 20, 30, and 40 jobs and a scheduling horizon set to 1.4 times the total processing time, which represents a moderate scheduling flexibility. The TOU tariff is varied by adjusting the magnitude of price fluctuations around the base level. In addition to the original setting with $\pm 70\%$ variation in mid-peak price, two additional scenarios with $\pm 30\%$ and $\pm 50\%$ variations are tested.

For each tariff scenario, we evaluate the energy consumption and compare the performance of the proposed integrated optimization model with the two-step strategy. This allows us to assess whether the advantages of the proposed method remain consistent under different pricing environments. The detailed results are presented in Table 6. The gap is calculated by $(EC'' - EC)/EC''$, where EC$''$ denotes the electricity cost obtained by using the two-step strategy. The subscript represents the variation in electricity prices.

Table 6. Results of the proposed model and the two-step strategy under different TOU tariffs.

Instance	$EC_{30\%}$	$EC''_{30\%}$	$Gap_{30\%}$	$EC_{50\%}$	$EC''_{50\%}$	$Gap_{50\%}$
20-1	2874.3	3070.9	6.40%	2276.5	2441	6.74%
20-2	3884.6	4039.4	3.83%	3039	3138	3.15%
20-3	3465.6	3563.6	2.75%	2732	2836	3.67%
20-4	3283.9	3356.7	2.17%	2532.5	2659	4.76%
20-5	3658.7	3731.6	1.95%	2898.5	3051	5.00%
20-6	4199.5	4283	1.95%	3353	3402	1.44%
20-7	2736.4	2826.6	3.19%	2146	2285.5	6.10%
20-8	3289.3	3379.5	2.67%	2495	2634.5	5.30%
20-9	3901.8	4028.3	3.14%	3028.5	3103	2.40%
20-10	3580.6	3735.8	4.15%	2745	2858.5	3.97%
Average			**3.22%**			**4.25%**
30-1	6404.9	6592.3	2.84%	5073	5228.5	2.97%
30-2	6607.1	6770.5	2.41%	5380.5	5618.5	4.24%
30-3	6439.8	6616.9	2.68%	5162.5	5377.5	4.00%
30-4	5094.6	5363.7	5.02%	4133	4450	7.12%
30-5	4340.4	4642.1	6.50%	3358	3839	12.53%
30-6	5357.9	5637.5	4.96%	4192	4483.5	6.50%
30-7	4515.7	4729.1	4.51%	3529	3876	8.95%
30-8	4815.3	5081.2	5.23%	3749.5	4150	9.65%
30-9	5212.8	5530.4	5.74%	4042	4428	8.72%
30-10	5944.2	6068	2.04%	4661	4928	5.42%
Average			**4.19%**			**7.01%**
40-1	9107.1	9446.1	3.59%	7409.0	7673.5	3.45%
40-2	*	7659.2	**	*	6129.0	**
40-3	8340.6	8589.8	2.90%	6923.0	7205.5	3.92%
40-4	6272.5	6504.1	3.56%	4932.5	5203.0	5.20%
40-5	6095.3	6549.6	6.94%	4928.5	5373.5	8.28%
40-6	6056.5	6471.9	6.42%	4703.5	5314.0	11.49%
40-7	7356.7	7675.6	4.15%	6017.0	6488.0	7.26%
40-8	6504.4	6812.4	4.52%	5233.5	5518.0	5.16%
40-9	6636.8	6948.0	4.48%	5299.5	5755.0	7.91%
40-10	7607.7	7893.8	3.62%	6142.0	6444.0	4.69%
Average			**4.46%**			**5.74%**

* Optimal solution is not obtained within 1800 s. ** Cannot obtain a gap due to lack of optimal solution.

By combining Tables 3 and 6, it can be observed that the smaller the fluctuation range of electricity prices, the greater the total electricity cost. This is because within a relatively flexible scheduling horizon, jobs are processed as much as possible during off-peak periods. The smaller the fluctuation range of electricity prices relative to mid-peak periods, the more expensive the electricity price is for off-peak periods, leading to an increase in total cost.

By combining Tables 5 and 6, it can be observed that the proposed integrated optimization model consistently outperforms the two-stage strategy across all tariff scenarios. The magnitude of the improvement is influenced by the degree of electricity price variation. In particular, as the magnitude of price fluctuation increases (from ±30% to ±50% and ±70%), the performance gap between the two approaches tends to increase. This suggests that the advantage of the proposed model becomes more pronounced under larger price variations. The results imply that larger peak–valley price differences amplify the benefit of coordinating scheduling and shutdown decisions.

Despite the variation in improvement magnitude, the overall performance trend remains consistent, indicating that the superiority of the integrated optimization framework is robust under different TOU tariff settings.

5. Conclusions

This paper studies a single-machine scheduling problem under time-of-use (TOU) electricity tariffs with machine switching decisions. A MILP model is developed to jointly optimize job scheduling and machine switching decisions, aiming to minimize total energy cost over a given planning horizon.

The computational results demonstrate that the proposed model can effectively exploit both load shifting and machine shutdown strategies to reduce energy consumption. Compared with the benchmark scenario without machine shutdown, substantial cost savings are consistently observed across all tested instances. Comparisons with a two-stage strategy further show that integrating scheduling and machine switching decisions leads to consistently better performance than sequential decision-making approaches. Sensitivity analysis results indicate that the performance advantage of the proposed model is affected by the level of electricity price variation, while the overall superiority of the integrated optimization framework remains robust. The magnitude of the energy-saving ratio depends on the level of scheduling flexibility and other parameter settings.

Furthermore, the results reveal that the scheduling horizon plays a critical role in determining the achievable cost reduction. As the horizon becomes longer, the model gains more flexibility to allocate jobs to low-price periods and to create sufficiently long idle intervals, which further enhances the benefits of machine switching decisions.

Overall, this study highlights the importance of integrating demand response mechanisms with machine-level operational strategies in production scheduling. Future research may extend this work by considering more complex machine environments, such as parallel or flow shop systems, and job features, such as release date and due date, incorporating uncertainty in electricity prices, or developing efficient solution algorithms for large-scale instances.

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Abbreviations

The following abbreviations are used in this manuscript:

CPP	Critical peak pricing
DR	Demand response
MILP	Mixed-integer linear programming
RTP	Real-time pricing
TOU	Time-of-use

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